1.9 NORMAL FORMS

1.9.1 Disjunctive Normal Form and Conjunctive Normal Form

- It is easy to observe that
1) A necessary and sufficient condition for an elementary product to be identically false (a contradiction) is that it contains atleast one pair of literals in which one is the negation of the other.

If $p \wedge \neg p \wedge \ldots$ is the formula, this reduces to $F \wedge \ldots$ and is equal to F.

2) A necessary and sufficient condition for an elementary sum to be identically true (a tautology) is that it contains atleast one pair of literals in which one is the negation of the other.

If $p \vee \neg p \vee \ldots$ is a formula, this reduces to $T \vee \ldots$ and is equal to T.

Definition 1: A formula which is equivalent to a given formula and consists of a sum of elementary products is called a disjunctive normal form (DNF).

To bring any formula to disjunctive normal form, first replace $\rightarrow$ and $\leftrightarrow$ using $\wedge$ and $\vee$ and $\neg$. Then use De Morgan's law wherever necessary.

Example 1: Obtain disjunctive normal forms of

1. $(p \rightarrow q) \wedge \neg q$
2. $\neg(p \wedge q) \leftrightarrow (p \vee q)$

1. $(p \rightarrow q) \wedge \neg q \equiv (\neg p \vee q) \wedge \neg q$ (Conditional-disjunction equivalence)
$\equiv \neg q \wedge (\neg p \vee q)$ (2nd Commutative Law)
$\equiv (\neg q \wedge \neg p) \vee (\neg q \wedge q)$ (2nd Distributive Law)
$\equiv (\neg q \wedge \neg p) \vee (q \wedge \neg q)$ (2nd Commutative Law)
$\equiv (\neg q \wedge \neg p) \vee F$ (2nd Negation Law)
$\equiv (\neg q \wedge \neg p)$ (2nd Identity Law)

Ans: $(\neg q \wedge \neg p) \vee (q \wedge \neg q)$

2. $\neg(p \wedge q) \leftrightarrow (p \vee q) \equiv (\neg(p \wedge q) \wedge (p \vee q)) \vee (\neg(\neg(p \wedge q)) \wedge \neg(p \vee q))$
(Section 1.8 Table 8, Equivalence 3)
$\equiv ((\neg p \vee \neg q) \wedge (p \vee q)) \vee ((p \wedge q) \wedge (\neg p \wedge \neg q))$
(Both De Morgan's Law and Double Negation Law)